MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: PLUG, STRAIGHT-THRU, HERMAPHRODITIC CONTACT FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-55302.

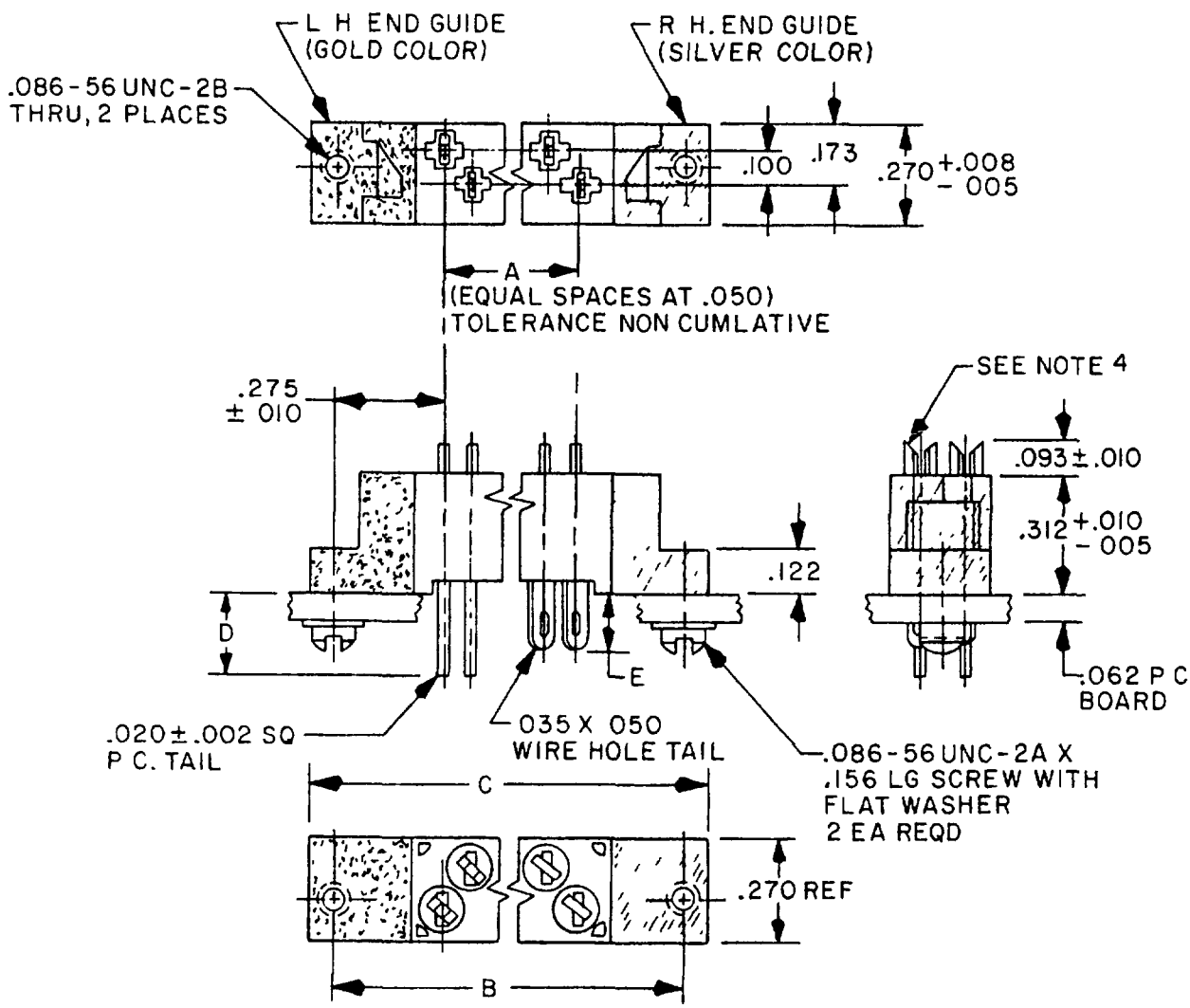
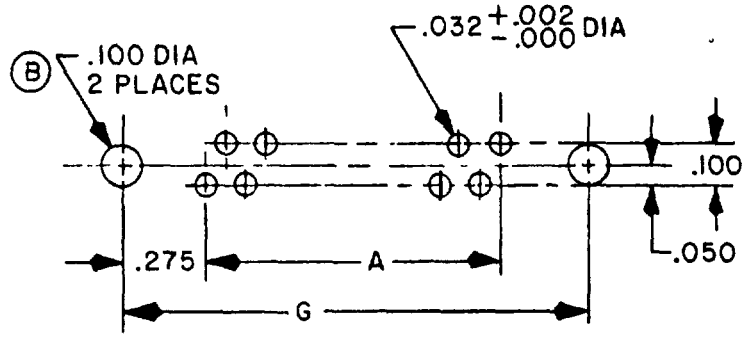


FIGURE 1. Connector, plug (.100 spacing).

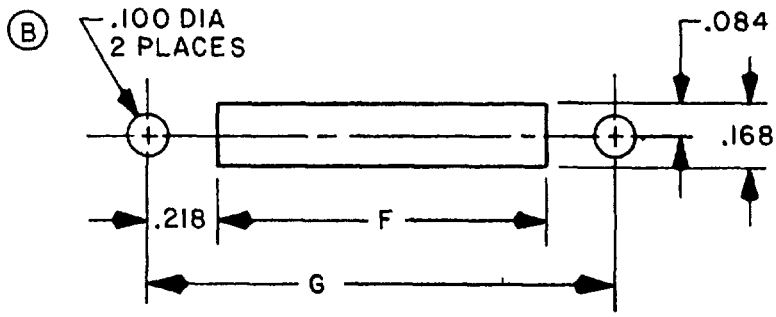
Ⓑ denotes changes

FSC 5935

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RECOMMENDED HOLE PATTERN FOR PLUG USING P C. TAIL CONTACTS.



RECOMMENDED CUTOUT FOR PLUG USING WIRE HOLE CONTACTS, IN NON-CONDUCTIVE CHASSIS

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.086	2.18	.270	6.86	1.750	44.45	2.764	70.21
.005	.13	.093	2.36	.275	6.98	1.864	47.35	3.200	81.28
.008	.20	.100	2.54	.312	7.92	2.000	50.80	3.340	84.84
.010	.25	.122	3.10	.453	11.51	2.050	52.07	3.550	90.17
.015	.38	.156	3.96	.531	13.49	2.140	54.36	3.664	93.07
.020	.51	.157	3.99	.850	21.59	2.164	54.97	4.100	104.14
.032	.81	.168	4.27	.964	24.49	2.300	58.42	4.240	107.70
.035	.89	.173	4.39	1.400	35.56	2.440	61.98		
.050	1.27	.218	5.54	1.450	36.83	2.600	66.04		
.062	1.57	.219	5.56	1.540	39.12	2.650	67.31		
.084	2.13	.250	6.35	1.564	39.73	2.740	69.50		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. With connectors fully engaged the mating contacts shall have a minimum overlap of .062. As a minimum the contacts shall exhibit a wiping action through this distance.

FIGURE 1. Connector, plug (.100 spacing) - Continued.

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TABLE I. Dash number and dimensions. (B)

Dash 1/ number	No. of contacts	Dimensions (inches)						
		A	B ±.010	C Ref.	D ±.015	E ±.015	F	G
01	18	.850	1.400	1.540	---	.157	.964	1.400
02					.219	---	---	
03					.250	---	---	
04					.531	---	---	
05					.093	---	---	
06					.453	---	---	
07	30	1.450	2.000	2.140	---	.157	1.564	2.000
08					.219	---	---	
09					.250	---	---	
10					.531	---	---	
11					.093	---	---	
12					.453	---	---	
13	36	1.750	2.300	2.440	---	.157	1.864	2.300
14					.219	---	---	
15					.250	---	---	
16					.531	---	---	
17					.093	---	---	
18					.453	---	---	
19	42	2.050	2.600	2.740	---	.157	2.164	2.600
20					.219	---	---	
21					.250	---	---	
22					.531	---	---	
23					.093	---	---	
24					.453	---	---	
25	54	2.650	3.200	3.340	---	.157	2.764	3.200
26					.219	---	---	
27					.250	---	---	
28					.531	---	---	
29					.093	---	---	
30					.453	---	---	
31	72	3.550	4.100	4.240	---	.157	3.664	4.100
32					.219	---	---	
33					.250	---	---	
34					.531	---	---	
35					.093	---	---	
36					.453	---	---	

1/ See requirements for complete part number when polarization is required.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material and finish:

(B) Contact. 0.20 inch thick phosphor bronze per QQ-B-750, composition "A". Gold plate per MIL-G-45204, type II, grade D, class 1, over nickel plating per QQ-N-290, class 2, 30 to 150 microinches in the engagement area for a length of .120 inch minimum (see figure 1). The remainder of the contact shall be gold plated per MIL-G-45204, type II, grade D, 10 microinches minimum over nickel plating per QQ-N-290, class 2, 30 to 150 microinches.

Insulator: MIL-M-14, type SDG-F.

End guides: Left hand, sintered brass 90/10, clear chromate per MIL-C-5541, class 1A (gold color). Right hand, sintered brass 90/10, nickel plate 300 microinches per MIL-F-14072 (silver color).

Polarizing pin: Aluminum alloy 2011-T3 with clear chromate, per MIL-C-5541, class 1A.

(B) Mounting screw: Brass, cadmium plated, gold chromate finish.

(B) Flat washer. Stainless steel.

Contact identification: Numerical sequence starting with the closest contact adjacent to the left hand end guide (gold color) shall be contact number 1; contact number 2 would be the next offset contact, etc.

Mating and unmating: The maximum insertion force, in pounds, shall not exceed a value equal to .5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of .11 times the number of contacts and shall not exceed the measured insertion force.

(B) Individual contact separation force:

Unplated contacts: One ounce minimum when tested in production with gage shown on MIL-C-55302/97.
Plated contacts: One ounce minimum.

Contact resistance: The contact resistance of all pairs measured shall not exceed 20 milliohms.

Low level circuit: The low level circuit resistance shall not exceed 20 milliohms.

Dielectric withstanding voltage.

Sea level: 1,000 volts rms, 60 cycles, ac.
High altitude (70,000 feet): 300 volts rms, 60 cycles, ac.

Current rating, maximum. 5 amperes.

Polarizing pin: When required, one of the following codes shall be added to the dash number:

"P" - specifying location by contact number where polarizing pin shall be inserted, for example, M55302/54-01P17 (polarizing pin inserted in place of contact number 17).
"H" - specifying location by contact number where contact shall be omitted for mating, for example, M55302/54-01H17 (polarizing hole has contact number 17 omitted).

(B) Mating receptacle: Shall conform to MIL-C-55302/52, /99, and /102.

Part number: M55302/54- (dash number from table I).

Patent number 2,750,572. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Custodians.
Army - CR
Navy - EC
Air Force - 17

Review activities.
Army - AR, MI
Navy - AS
Air Force - 11, 99
DLA - ES

User activities.
Army - AT, AV
Navy - MC, OS, SH
Air Force - 19

Preparing activity.
Army - CR

Agent:
DLA - ES

(Project 5935-3145-3)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-55302/54B Connectors, Printed Circut Subassembl and Accessories: Plug, Straight-thru, Hermaphroditic Contact fpr printed Wiring Boards (.100) Spacing

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

HQ, USA CORADCOM
ATTN: DRDCO-CM-DM
Ft. Monmouth, N.J., 07703

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